

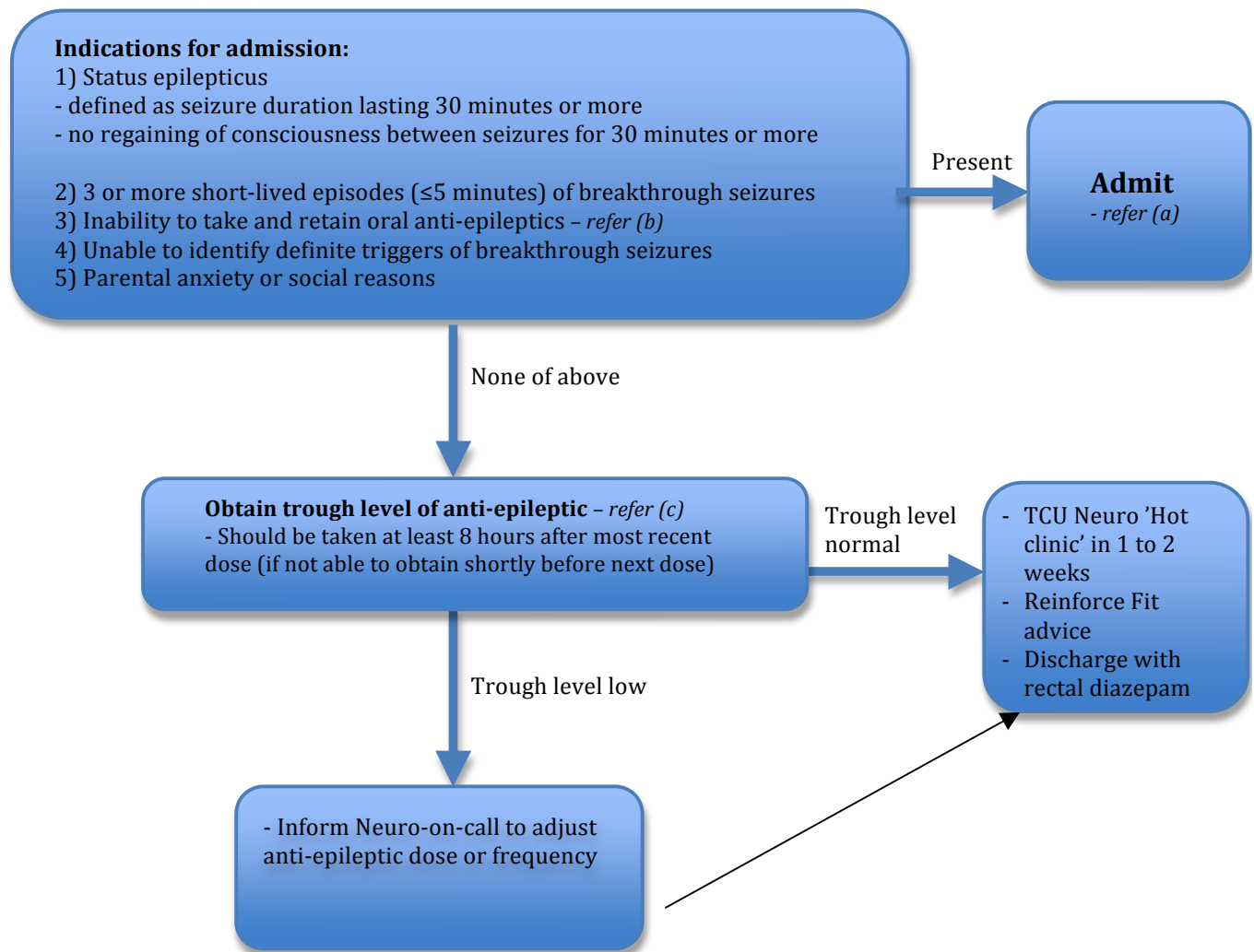
Guidelines for the Management and Disposition of Breakthrough Seizure in Children's Emergency

This document presents guidelines for the management and disposition of paediatric patients (<16 years old) with known epilepsy, who present to our Children's Emergency following breakthrough seizure/s.

A breakthrough seizure is defined as an epileptic event that occurs in a patient with diagnosed epilepsy, despite the use of long-term anticonvulsants.

Common triggers or precipitating factors for breakthrough seizures:

- Suboptimal compliance with antiepileptic medications
- Sleep deprivation
- Physical exertion
- Emotional stress
- Watching television or playing video games
- Fever/ concurrent illness
- Decreased anti-epileptic levels due to drug interactions
- Electrolyte abnormalities, hypoglycemia



(a) On admission

- Electrolytes and blood glucose, especially in the context of decreased oral intake or intercurrent infection
- Trough level of anti-epileptic which should be taken at least 8 hours after most recent dose; if not able to obtain shortly before next dose (refer to (c))
- May need to give loading dose of anti-epileptic if levels are low (consult neuro on-call)

(b) Anti-epileptics that can be converted to IV

- Phenytoin
- Sodium Valproate (Epilim)
- Levetiracetam (Keppra)
- Diazepam

(c) Reference ranges of common anti-epileptics

*circled – available in NUH lab

Table 1. The “Reference Range” for Commonly Prescribed AEDs		
AED	“Usual” Daily Dose	Reference Range
carbamazepine	400-1800+ mg	4-12 ug/ml
ethosuximide	500-1500+ mg	40-100 ug/ml
felbamate	1200-4800+ mg	30-100 ug/ml
gabapentin	900-3600+ mg	4-20 ug/ml
lamotrigine	200-800+ mg	4-20 ug/ml
levetiracetam	1000-4000+ mg	5-40 ug/ml
oxcarbazepine	600-2700+ mg	10-40 ug/ml
phenytoin	200-400+ mg	10-20 ug/m
phenobarbital	90-180+ mg	20-40 ug/ml
tiagabine	12-64+ mg	100-300 ng/ml
topiramate	100-600+ mg	10-20 ug/ml
valproate	750-2500+ mg	50-100 ug/m
zonisamide	100-600+ mg	10-40 ug/ml

References:

1. Steven Karciski. Don't Get Fooled by Misleading Serum Levels of Antiepileptic Medications. *Practical Neurology* August 2005: 60-61
2. Table of Antiepileptic Drugs. Published by Tuberous Sclerosis Alliance
www.tsalliance.org